DeepSeek Novelty Judging

Replication progress and preliminary findings

Research purpose and design

Test whether the sensitivity of LLM novelty judgments extends to a Chinese-model judge. We retain the authors' English benchmark and change the judge to DeepSeek. The data include Human-Only and Human+Generated settings; this is not a wholly Chinese-model data-generation pipeline.

Pointwise evaluation asks whether one idea is novel (299 / 308 items); pairwise evaluation chooses between two ideas (154 pairs per setting). The legacy reference uses three pointwise votes or three votes per presentation order for pairwise evaluation. Prompt wording, aggregation, input format and pre-generated negative sources are varied.

Verified results

40 completed configurations pass item-index, label, vote-count and aggregation checks. The original 37 configurations and three Human-Only plan-prompt supplements are complete. The authors' three-subrun procedure yields 22 no-aggregation conditions, covering 18 additional conditions after deduplication (58 conditions in total).

Setting	Pointwise macro-F1	Pairwise strict	Pairwise soft	Tie rate
Human-Only	0.733	0.812	0.838	5.2%
Human+Generated	0.786	0.662	0.705	8.4%

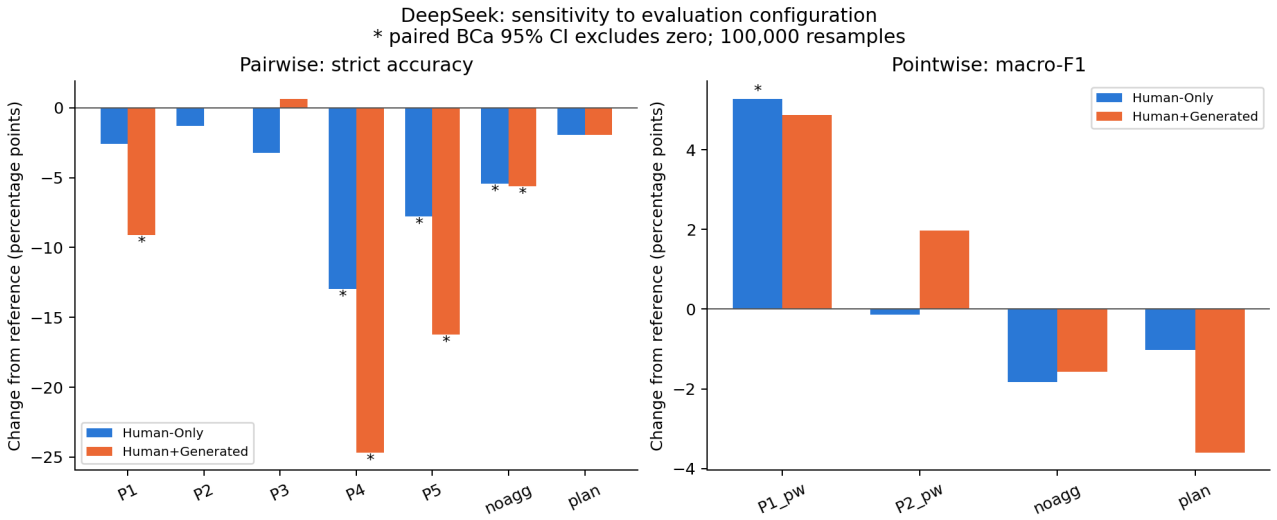


Figure 1. Changes from the reference. Stars indicate paired BCa 95% intervals excluding zero; 100,000 resamples per comparison, without multiple-testing correction.

On Human+Generated data, P4 reduces strict accuracy by **24.68 percentage points** (95% CI: -32.47 to -17.53). This supports prompt sensitivity for this judge and benchmark; it does not establish the true novelty of individual ideas.

Source effects and progress against the paper

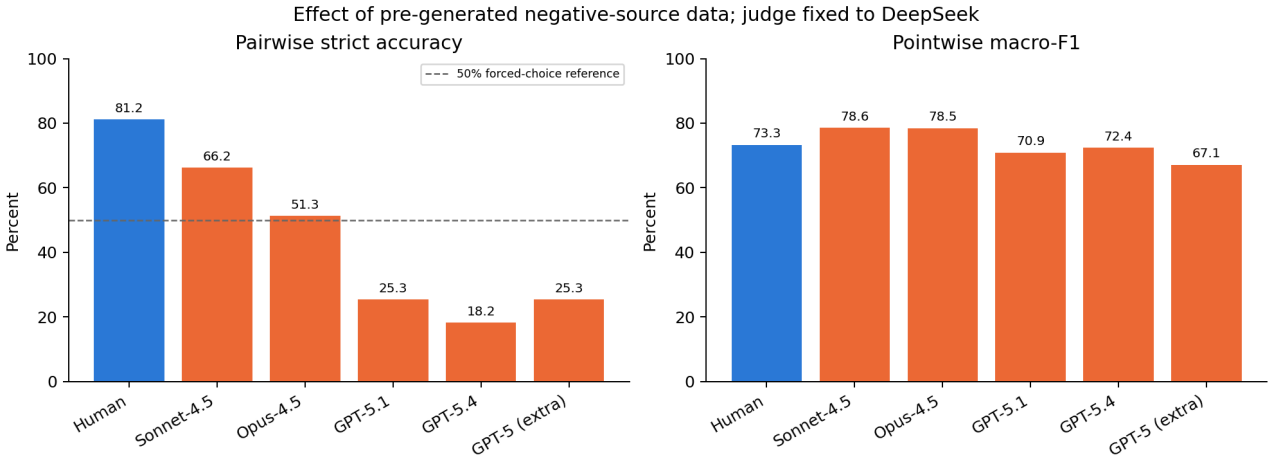


Figure 2. The judge stays DeepSeek; only the pre-generated negative-source dataset changes. GPT-5 is an extra source beyond the paper's listed comparison. No foreign-model API was called to obtain these archived data.

Paper component	Current coverage / remaining work
Fig. 4: controlled changes	26 of 34 single-judge configurations. Retrieval and explicit low-effort conditions remain.
Fig. 12: plans + P3/P4/P5	6 of 6 configurations; both dataset settings complete.
Fig. 13: generator sources	All three additional sources listed in the paper, both protocols, complete.
Fig. 11: no aggregation	18 of 26 supplementary conditions derived from three k=1 subruns; 8 retrieval/low-effort supplements remain.
Figs. 5, 6/7 and 14	Agentic retrieval, dedicated evaluators and cross-source retrieval remain.

Current problems and next steps

Reasoning control. Historical calls used a model alias and did not explicitly lock thinking effort. Current low/high probes work, but do not establish historical behavior. Matched explicit-high references are needed before interpreting low-effort effects.

Retrieval fidelity. The paper uses contribution extraction, five targeted queries, Paper Finder and a 2025-03-01 cutoff. A generic web-search substitute would be a changed protocol. Dedicated evaluators require separate implementation and tool-cost measurement.

Evidence limits. Labels and expert validation are inherited. Some historical raw responses are unavailable, and parsers changed across stages. Strict accuracy counts ties as zero; values below 0.5 alone do not prove systematic reversed judgments.

Analysis archive: 120 paired BCa comparisons. Paper-aligned derivations and supplementary fresh repeats are stored separately. Extra fresh calls were stopped after checking the authors' derivation procedure. This is a selected single-judge replication, not the complete multi-model paper.

Source: Sternlicht et al., Old Ideas, Novel Problems: The Instability of LLM-Based Novelty Evaluation, arXiv:2610.02022v1. Benchmark: huggingface.co/datasets/noystl/novelty-judge-bench. Code: github.com/Viego051115/novelty-eval-replication.